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(19) **United States**(12) **Patent Application Publication** (10) **Pub. No.: US 2025/0280955 A1**  
Murphy (43) **Pub. Date: Sep. 11, 2025**(54) **RECLINING SEATING UNIT FOR USE IN THEATER WITH RISER ARRANGEMENT**(71) Applicant: **Ultra-Mek, Inc.**, Denton, NC (US)(72) Inventor: **Marcus L. Murphy**, Lexington, NC (US)(21) Appl. No.: **19/218,772**(22) Filed: **May 27, 2025****Related U.S. Application Data**

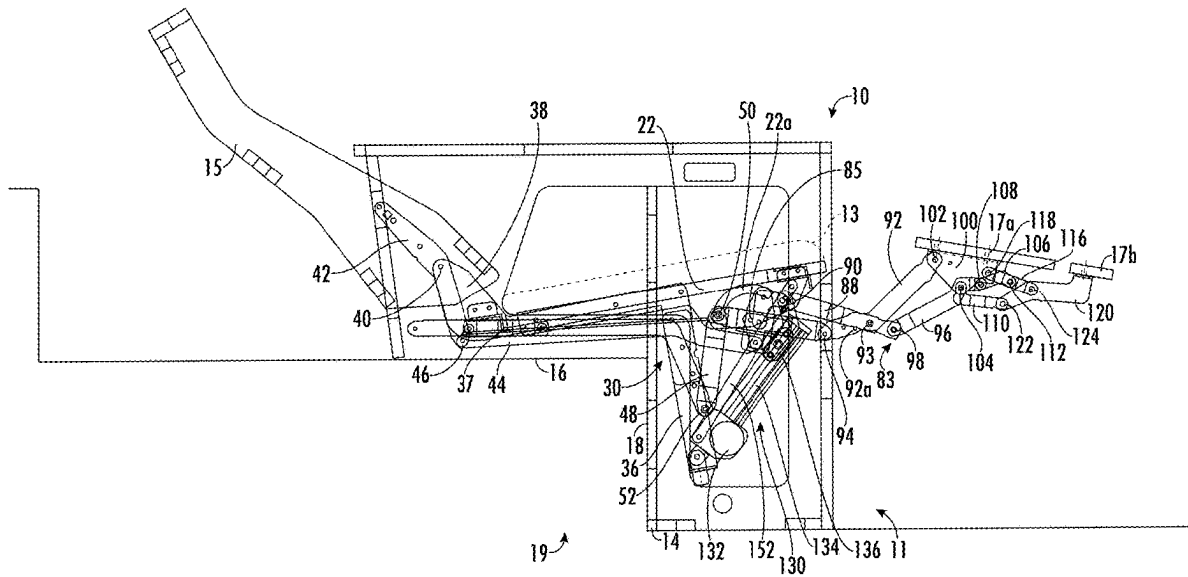
(63) Continuation-in-part of application No. 18/604,615, filed on Mar. 14, 2024.

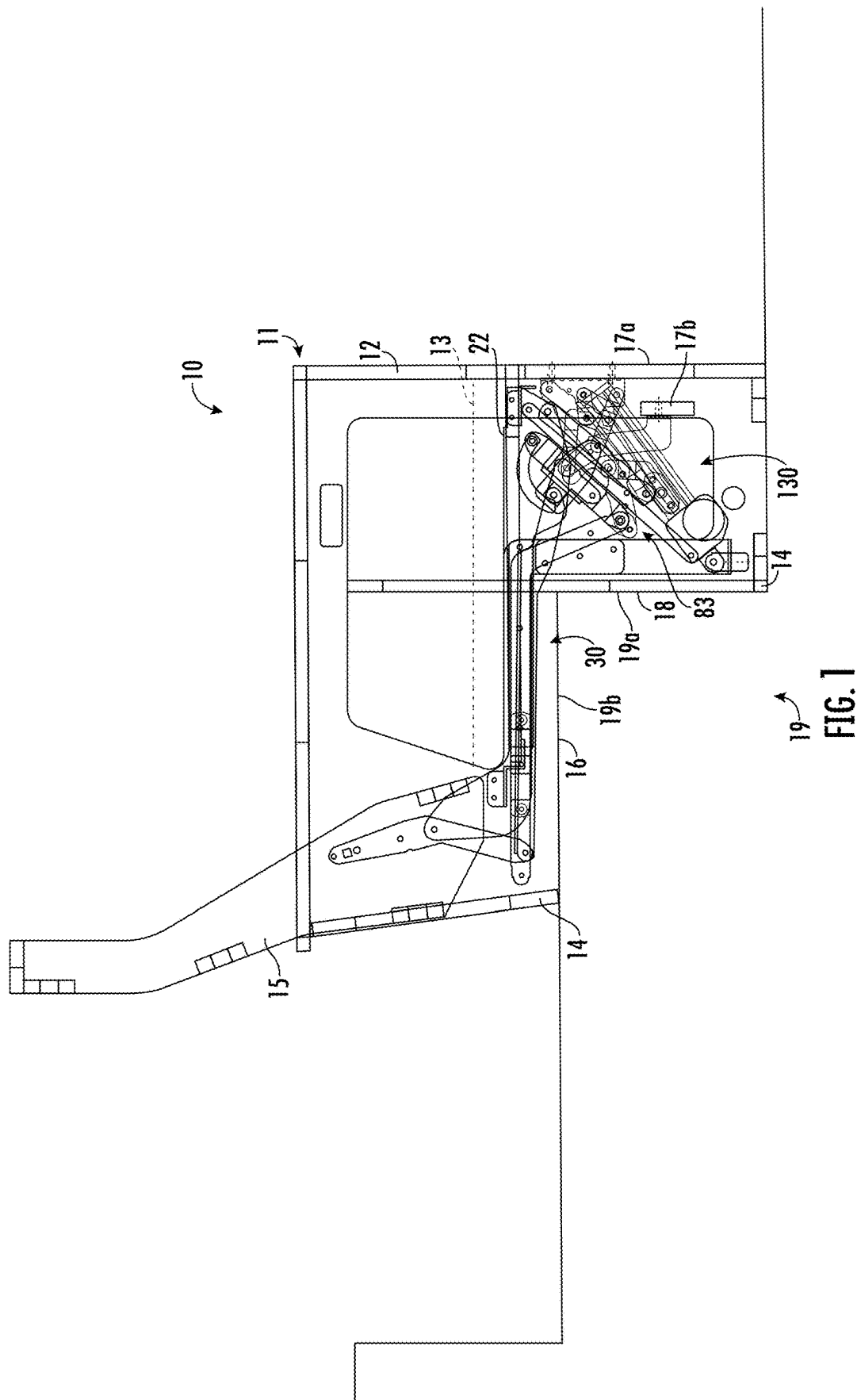
(60) Provisional application No. 63/459,027, filed on Apr. 13, 2023.

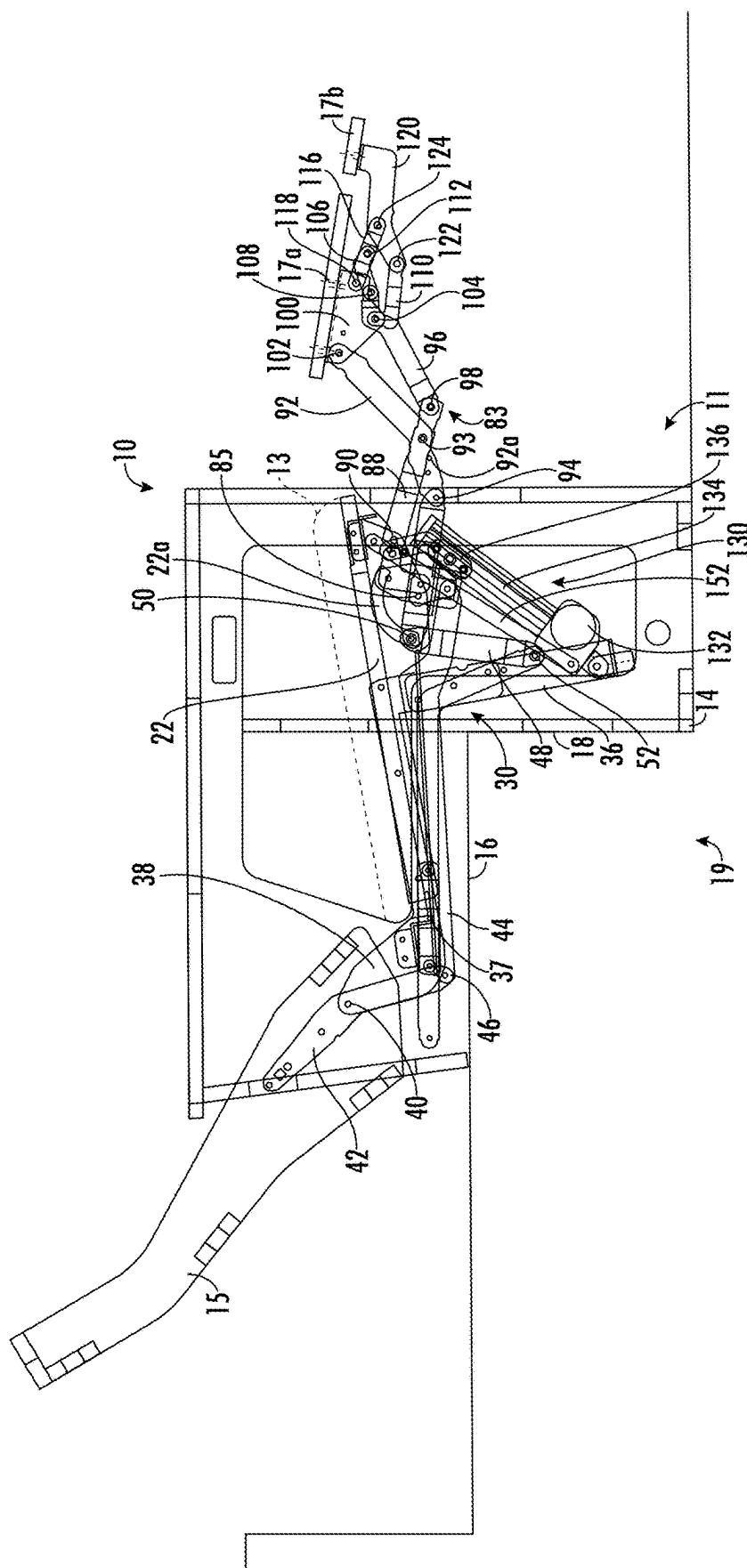
**Publication Classification**(51) **Int. Cl.***A47C 1/0355* (2013.01)*A47C 1/035* (2006.01)(52) **U.S. Cl.**CPC ..... *A47C 1/0355* (2013.01); *A47C 1/0352* (2013.01)(57) **ABSTRACT**

A seating unit includes: a frame having a pair of arms, wherein each of the arms includes a cutaway portion in a

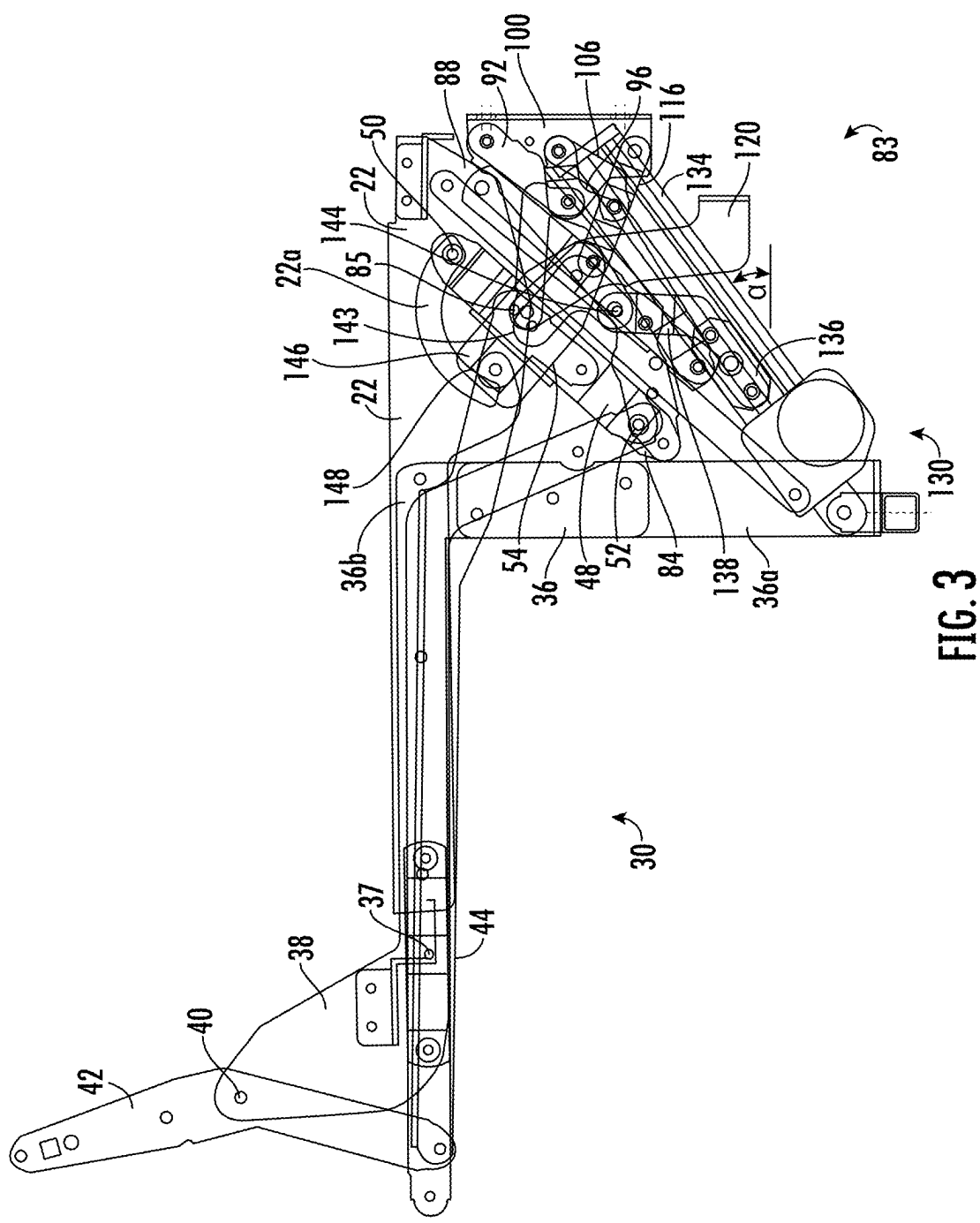
lower rear area thereof, the cutaway portion configured to receive a tread and a riser of a riser arrangement; a backrest; a seat; a first footrest; a reclining mechanism; and a linear actuator. The reclining mechanism is connected between the backrest, seat, and first footrest, the reclining mechanism comprising a series of pivotally interconnected links and configured to move the seating unit between: (a) an upright position, in which the backrest is disposed at a first generally upright backrest angle relative to the seat, the seat is disposed at a first generally horizontal seat angle and at a first longitudinal position, and the first footrest is retracted below a forward portion of the seat; and (b) a fully reclined position, in which the backrest is disposed at a second backrest angle relative to horizontal that is shallower than the first backrest angle, the first footrest is extended in front of the seat, the seat has a second seat angle that is deeper relative to horizontal than the first seat angle, and the seat has a second longitudinal position that is rearward of the first longitudinal position. The reclining mechanism includes a footrest mechanism attached to the first footrest, the footrest mechanism configured to control relative movement of the first footrest relative to the seat. The linear actuator is attached to the reclining mechanism and the footrest linkage and configured to move the backrest, seat and first footrest between the upright and fully reclined positions. The linear actuator and the footrest linkage reside entirely below a front portion of the seat and in front of the cutaway portion of the arms.







**FIG. 2**



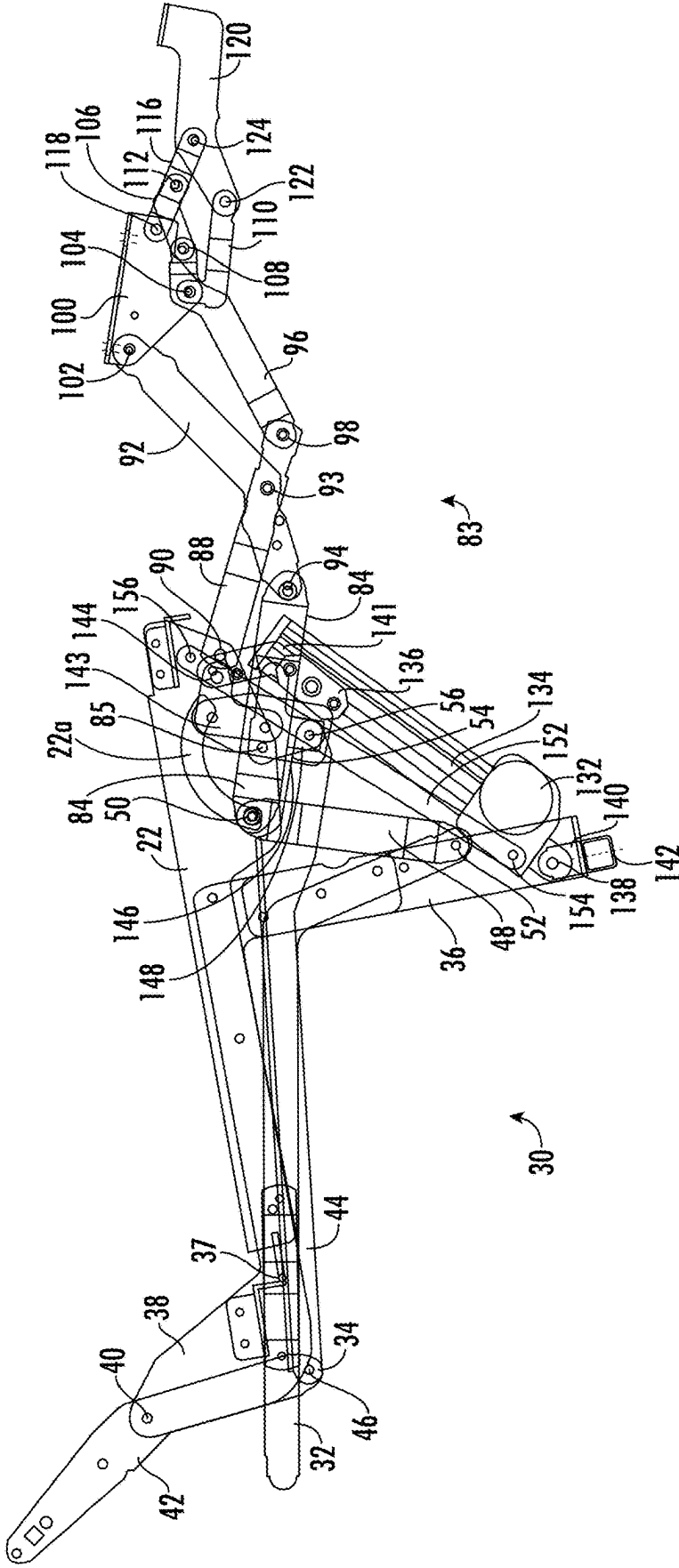


FIG. 4

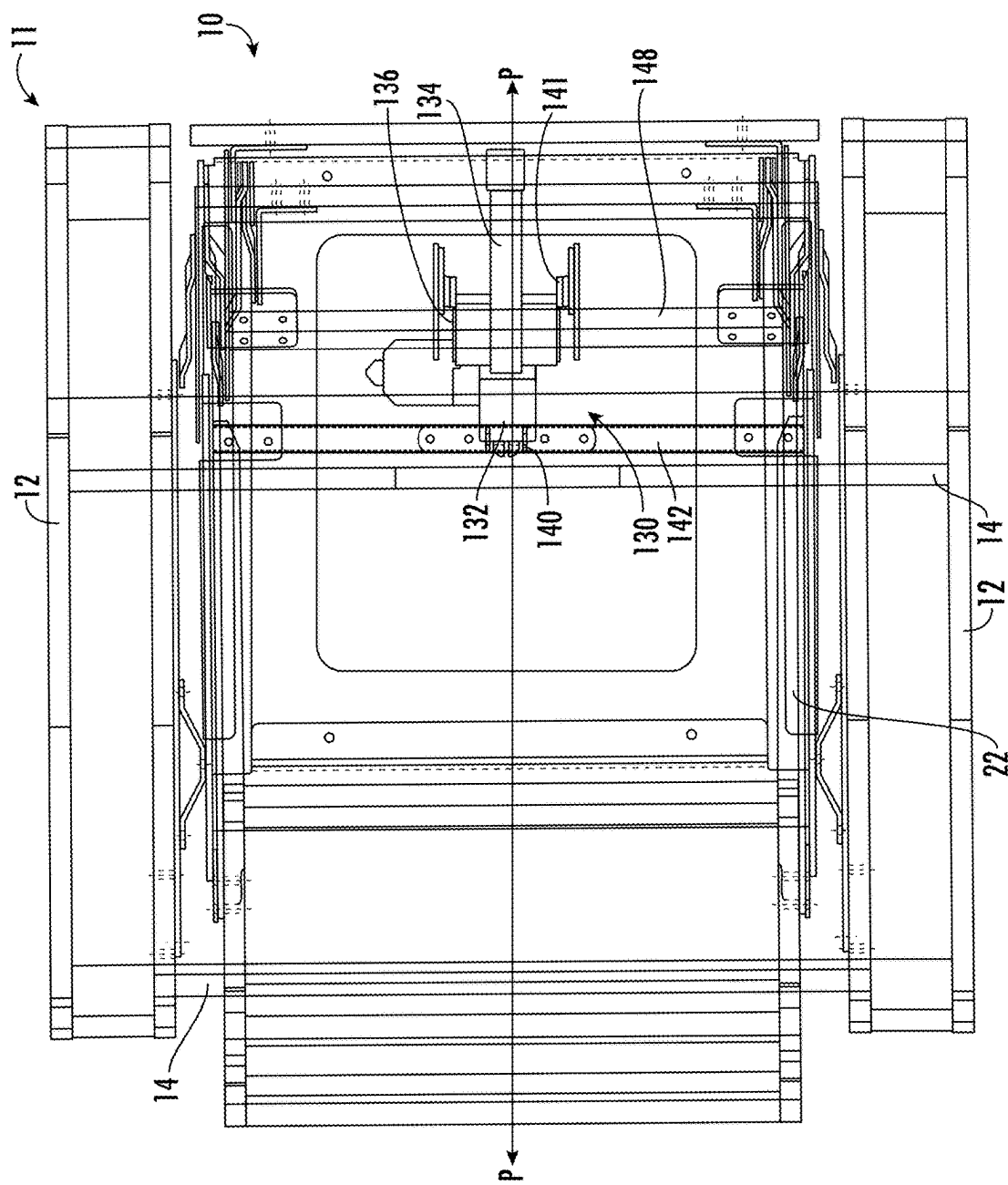


FIG. 5

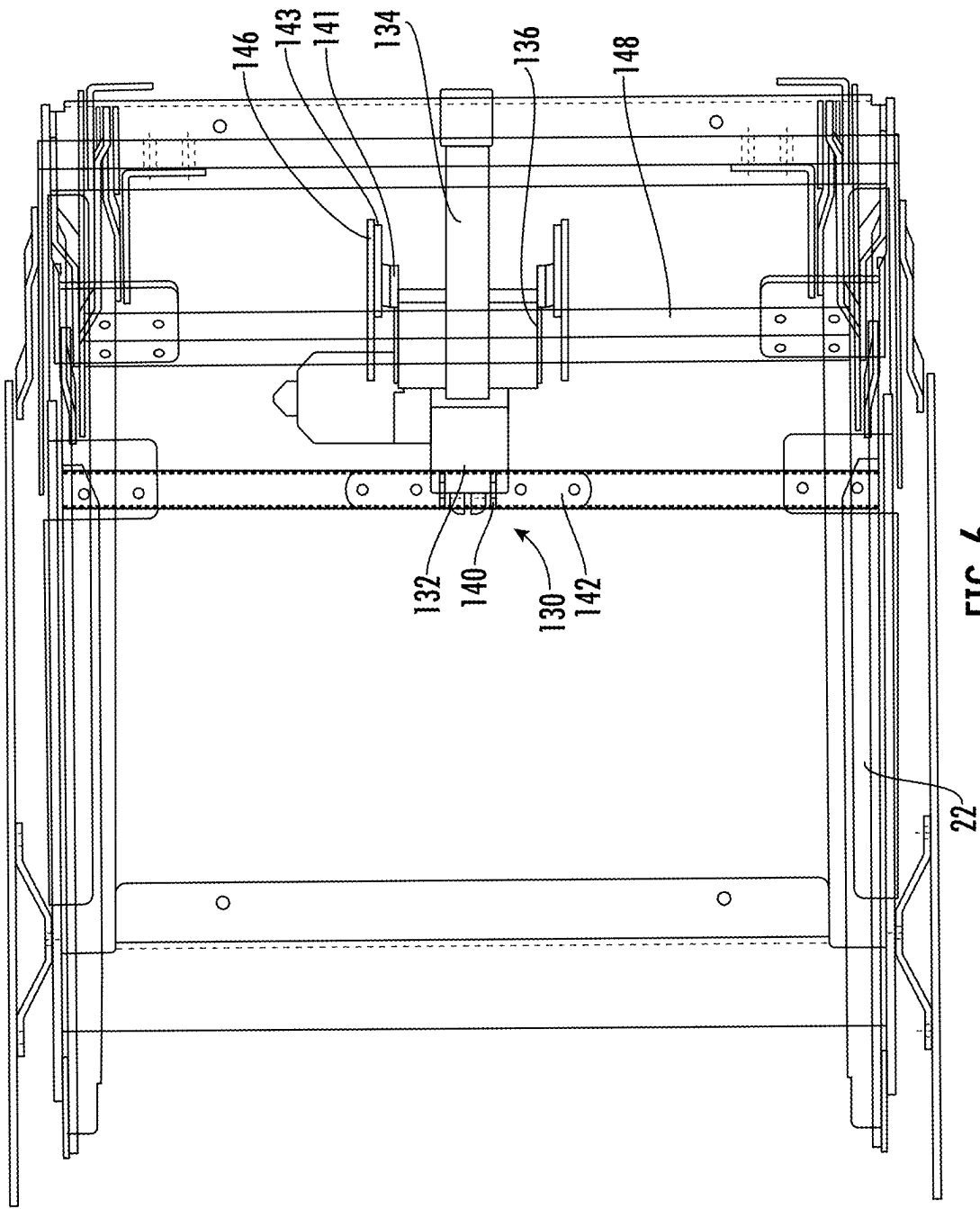
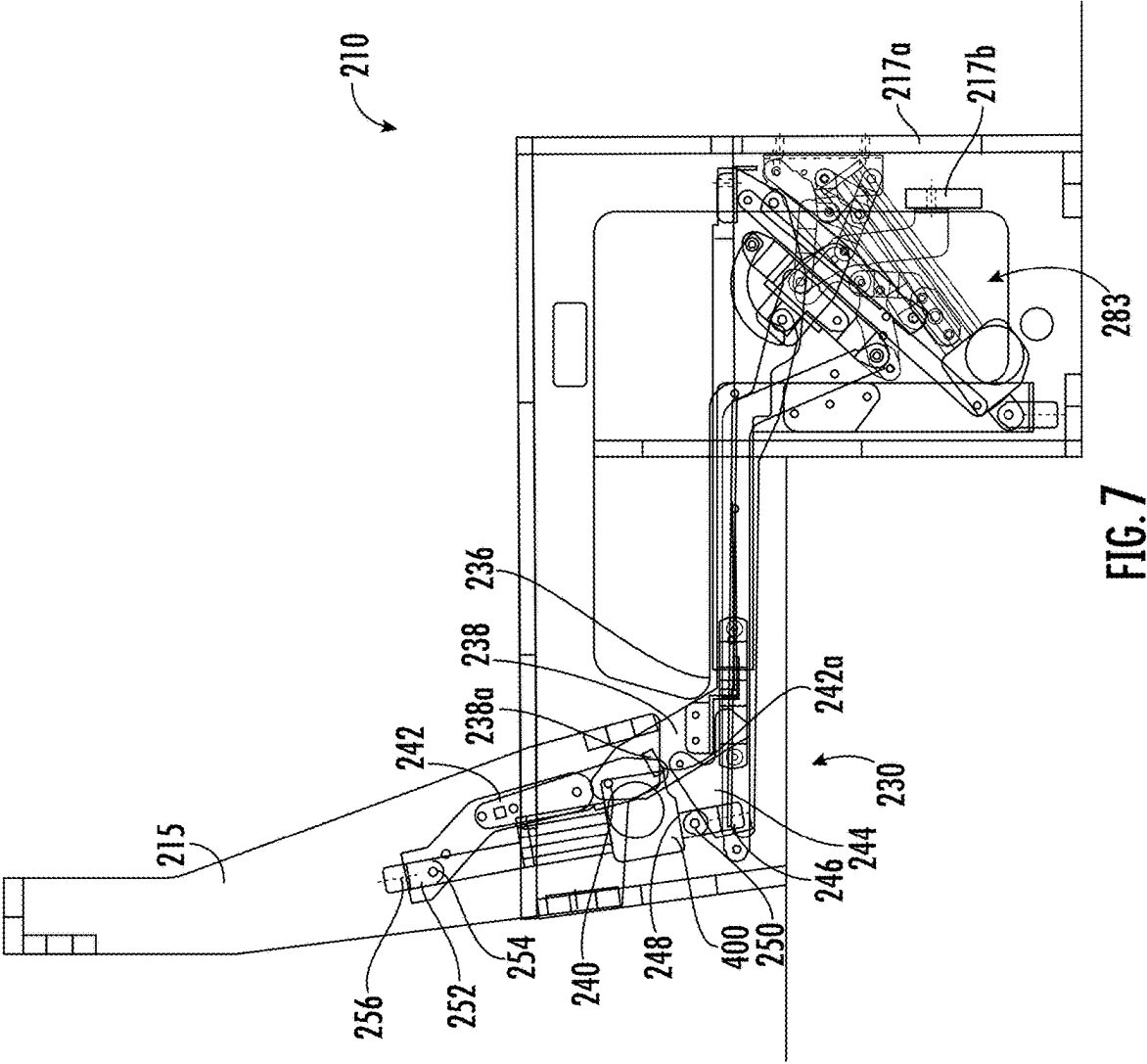
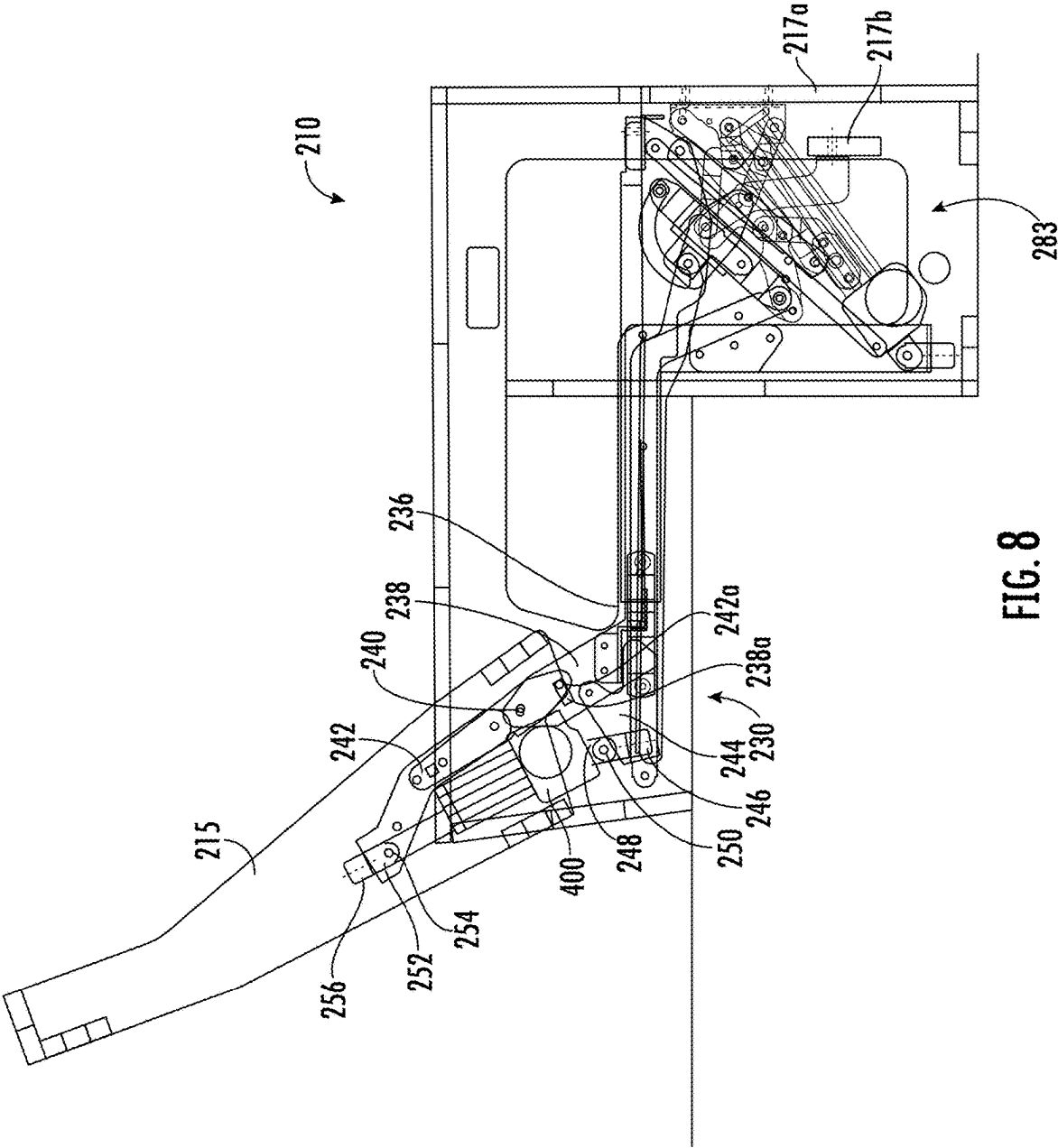


FIG. 6







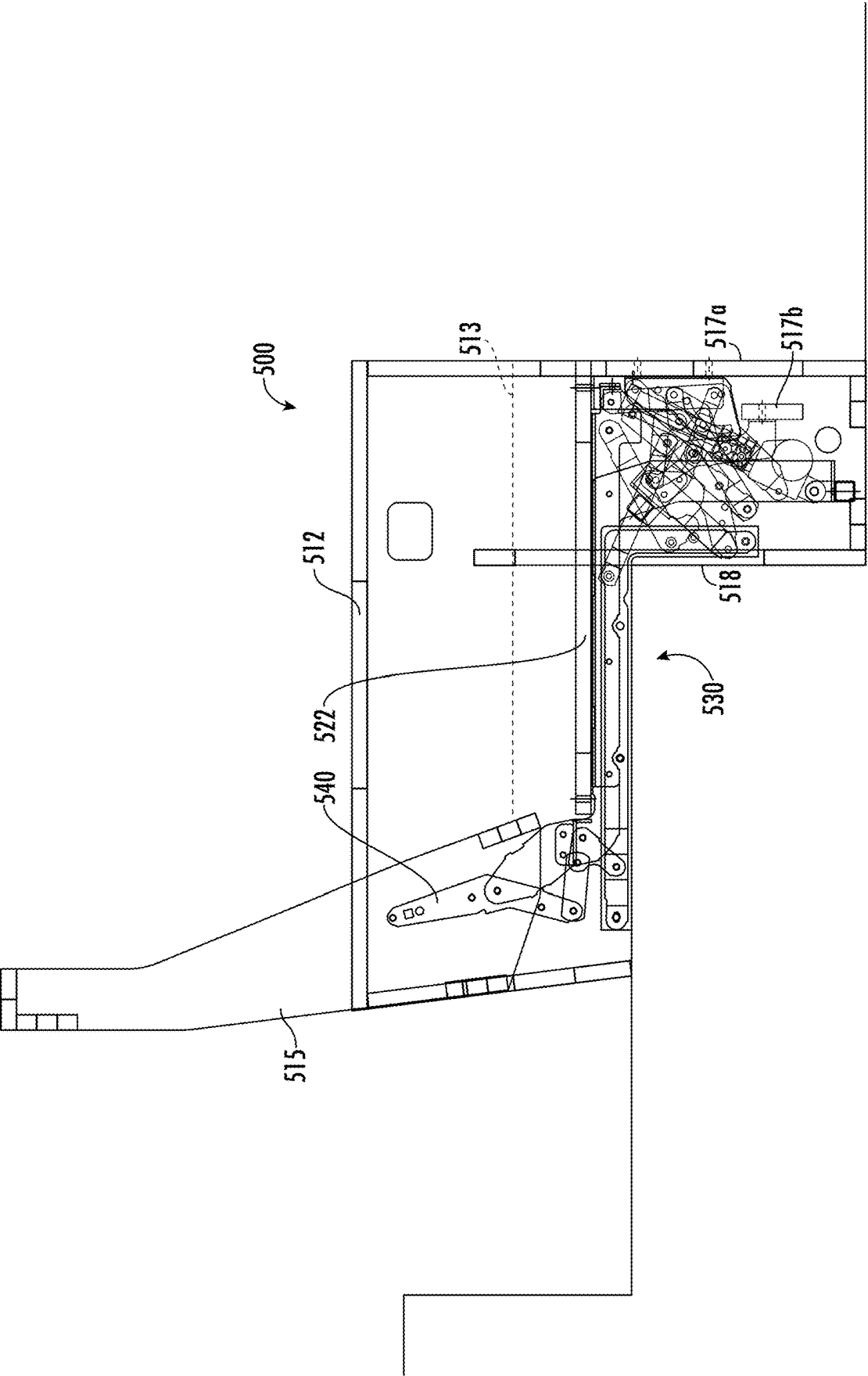


FIG. 9

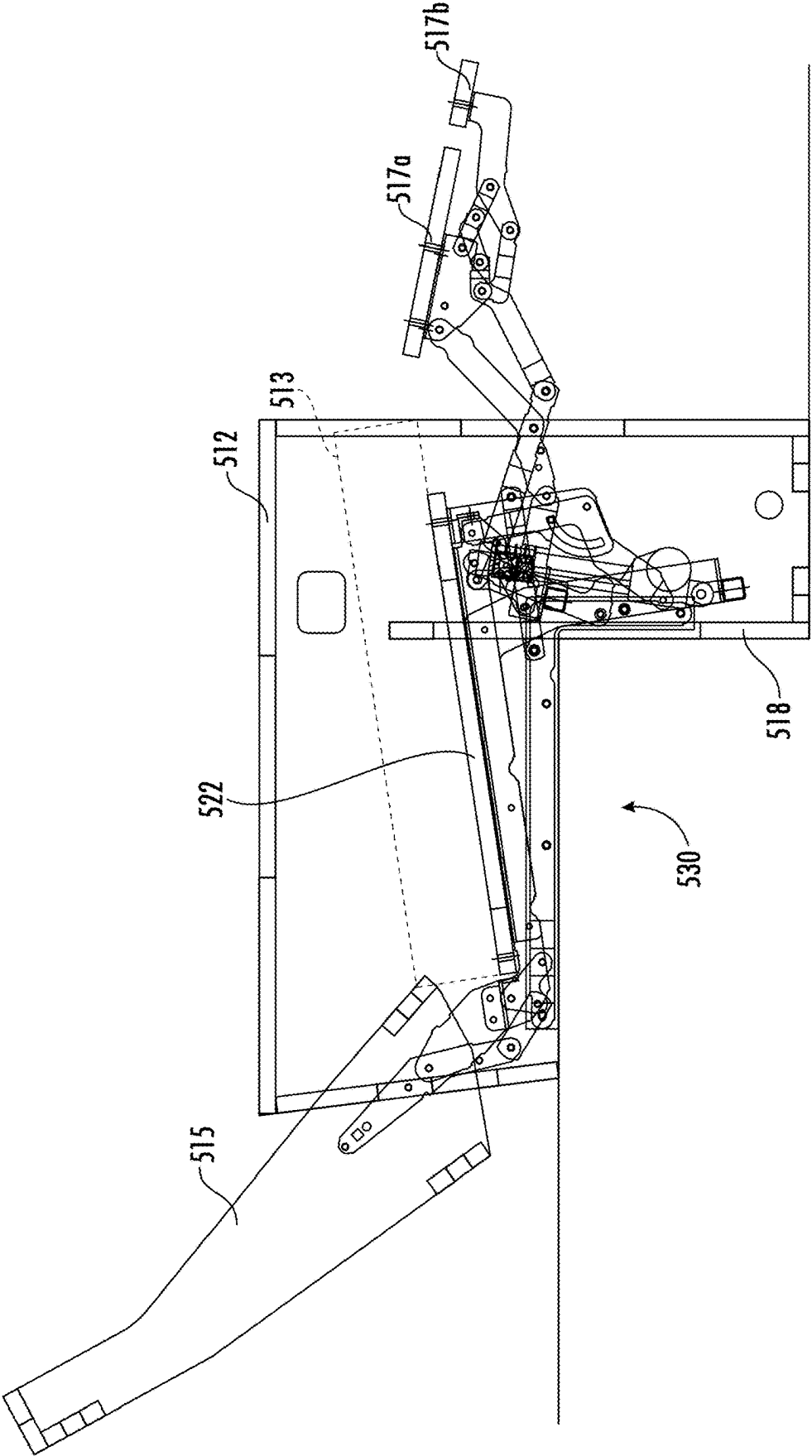
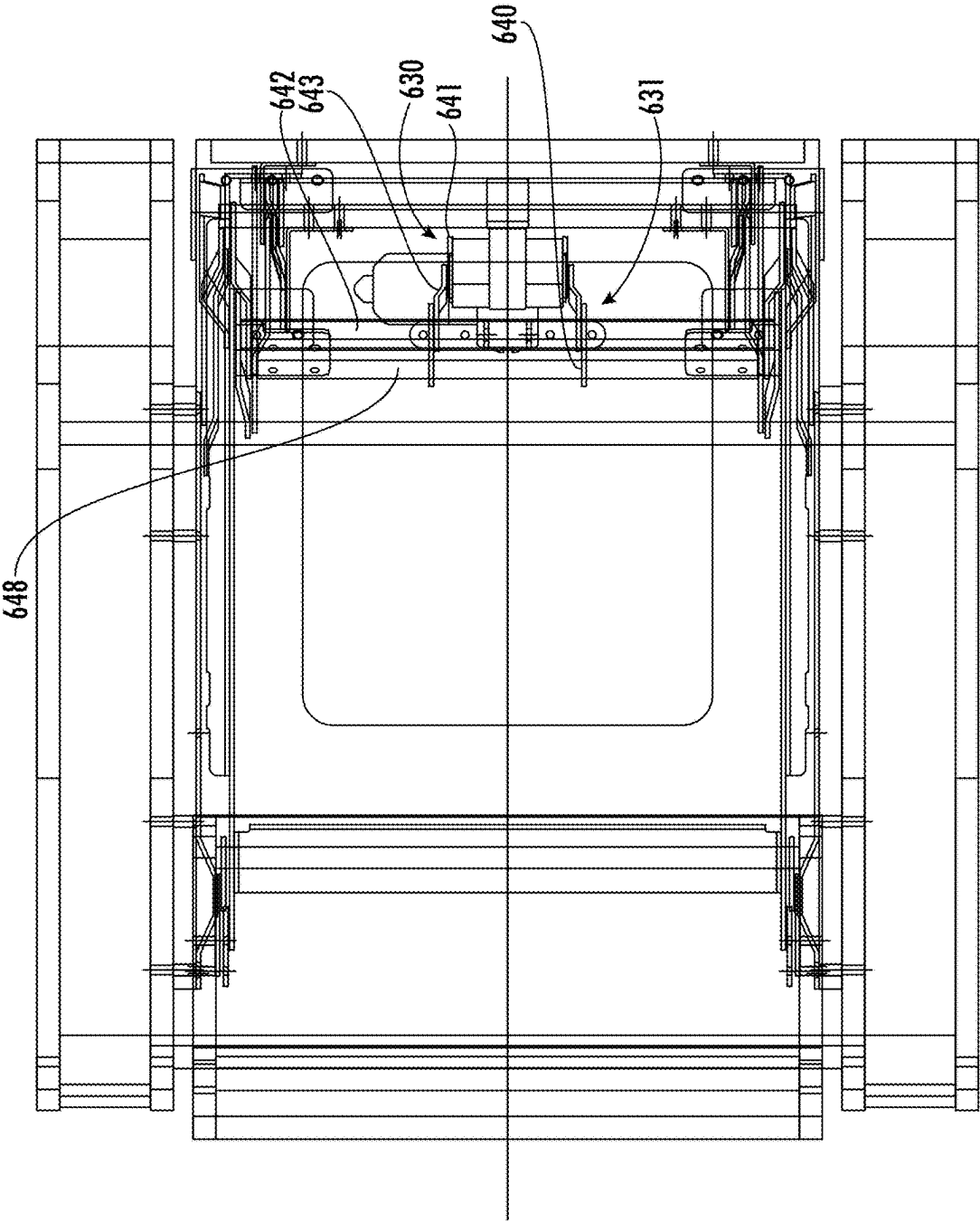
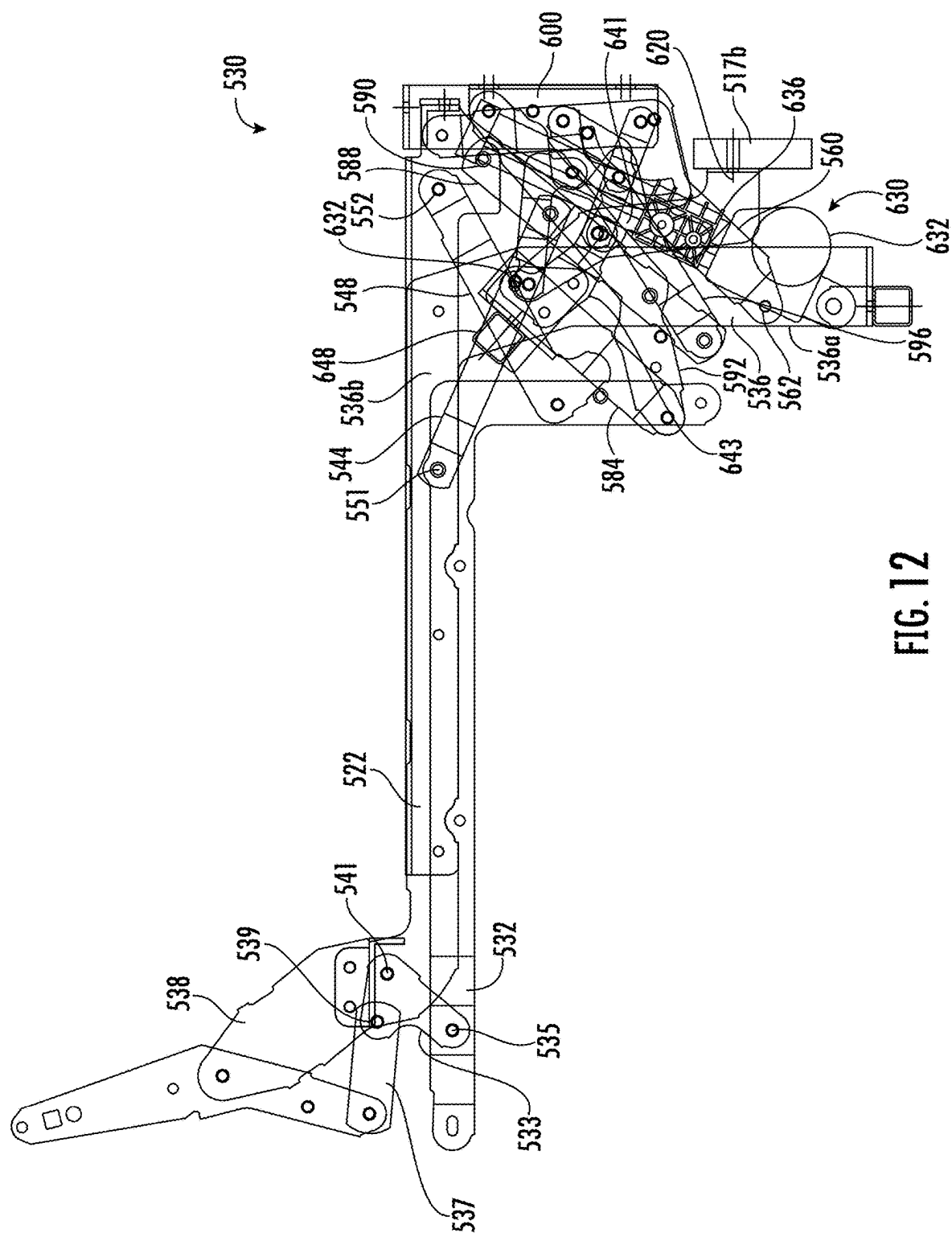


FIG. 10





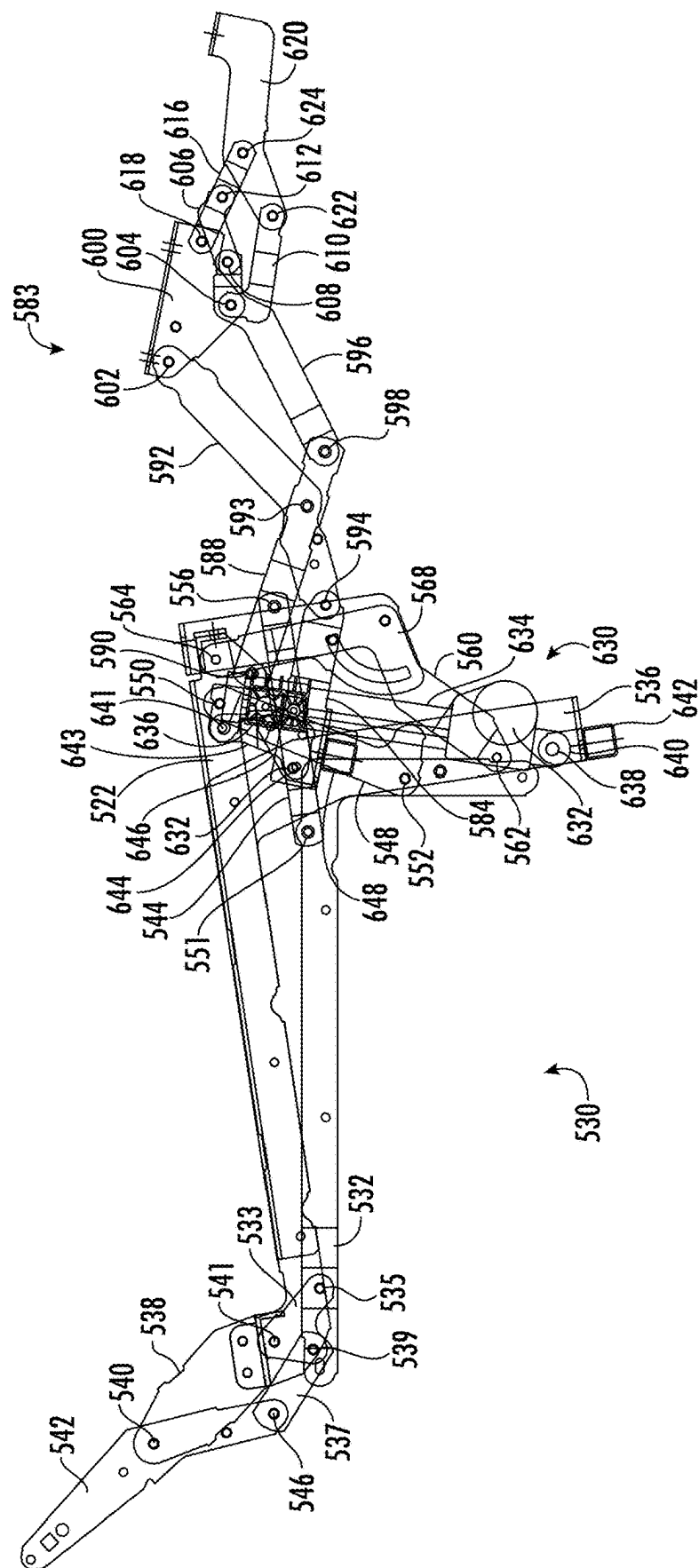


FIG. 13

## RECLINING SEATING UNIT FOR USE IN THEATER WITH RISER ARRANGEMENT

### RELATED APPLICATIONS

[0001] This application is a continuation-in-part of and claims priority to U.S. patent application Ser. No. 18/604,615, filed Mar. 24, 2024, which claims the benefit of U.S. Provisional Patent Application No. 63/459,027, filed Apr. 13, 2023, the disclosure of each of which is hereby incorporated therein by reference in full.

### FIELD OF THE INVENTION

[0002] The present invention relates generally to seating units, and relates more particularly to reclining seating units.

### BACKGROUND OF THE INVENTION

[0003] Recliner chairs and other reclining seating units have proven to be popular with consumers. These seating units typically move from an upright position, in which the backrest is generally upright, to one or more reclined positions, in which the backrest pivots to be less upright. The movement of the seating unit between the upright and reclined positions is typically controlled by a pair of matching reclining mechanisms that are attached to the seat, backrest and base of the chair.

[0004] One particularly popular recliner is the “three-way” recliner, which has two reclined positions: a “TV position”, in which the footrest or ottoman of the chair is projected forwardly from the chair while the backrest remains substantially upright; and a “fully reclined position”, in which the backrest is less upright (i.e., it has been reclined to a shallower angle relative to the floor). In a “three-way” recliner, the backrest pivots relative to the seat as the chair takes its fully reclined position; this differs from a “two-way” recliner, in which the backrest and seat are rigidly fixed and do not pivot relative to one another as the chair moves to the fully reclined position. Many three-way recliners are constructed such that the backrest and footrest are coupled to one another, such that reclining of the backrest cannot occur unless the footrest is already extended (i.e., the chair is in the TV position). See, e.g., U.S. Pat. No. 4,915,444 to Rogers, Jr. and U.S. Pat. No. 6,540,291 to Hoffman, which illustrate chairs of rather contemporary style with three-way reclining capability. Other reclining chairs may have decoupled reclining and footrest-extending mechanisms. See, e.g., U.S. Pat. No. 7,731,276 to Hoffman et al. Some reclining units have employed power actuators to recline the backrest and extend the footrest. See, e.g., U.S. Pat. No. 8,297,693 to Hoffman et al.

[0005] With the ubiquity and variety of reclining seating units (and their accompanying reclining mechanisms), reclining seating units now appear in many locations outside the home. One place that has been popular for the installation of reclining seating units is movie theaters, particularly those trying to compete with “home theater” experiences. However, retrofitting existing theaters with reclining seats can be challenging because of the presence of riser arrangements within the theaters, which provide a platform or foundation for theater seats that place the seats at the rear of the theater at a higher elevation than the seats in the front. Riser arrangements are typically configured similar to the steps of a staircase, with alternating horizontal “treads” and vertical “risers” that form the “steps.” Typical conventional

theater seats are mounted on the treads. However, if a reclining seating unit is to be installed, typically a portion of the tread and riser must be removed in order to have room for a reclining seating unit and its reclining mechanism. This can be very expensive (often prohibitively expensive), thus putting the existing theater at a competitive disadvantage to newer theaters. Thus, it may be desirable to provide a reclining seating unit that can be retrofitted into a theater with a conventional riser arrangement.

### SUMMARY

[0006] As a first aspect, embodiments of the invention are directed to a reclining seating unit. The reclining seating unit comprises: a frame having a pair of arms, wherein each of the arms includes a cutaway portion in a lower rear area thereof, the cutaway portion configured to receive a tread and a riser of a riser arrangement; a backrest; a seat; a first footrest; and a reclining mechanism connected between the backrest, seat, and first footrest. The reclining mechanism comprises a series of pivotally interconnected links and is configured to move the seating unit between: (a) an upright position, in which the backrest is disposed at a first generally upright backrest angle relative to the seat, the seat is disposed at a first generally horizontal seat angle and at a first longitudinal position, and the first footrest is retracted below a forward portion of the seat; and (b) a fully reclined position, in which the backrest is disposed at a second backrest angle relative to horizontal that is shallower than the first backrest angle, the first footrest is extended in front of the seat, the seat has a second seat angle that is deeper relative to horizontal than the first seat angle, and the seat has a second longitudinal position that is rearward of the first longitudinal position. The reclining mechanism includes a footrest mechanism attached to the first footrest, the footrest mechanism configured to control relative movement of the first footrest relative to the seat. The reclining seating unit further comprises a linear actuator attached to the reclining mechanism and the footrest linkage and configured to move the backrest, seat and first footrest between the upright and fully reclined positions. The linear actuator and the footrest linkage reside entirely below a front portion of the seat and in front of the cutaway portion of the arms.

[0007] As a second aspect, embodiments of the invention are directed to a reclining seating unit comprising: a frame having a pair of arms; a backrest; a seat; a first footrest; and a reclining mechanism connected between the backrest, seat, and first footrest. The reclining mechanism comprises a series of pivotally interconnected links and is configured to move the seating unit between: (a) an upright position, in which the backrest is disposed at a first generally upright backrest angle relative to the seat, the seat is disposed at a first generally horizontal seat angle and at a first longitudinal position, and the first footrest is retracted below a forward portion of the seat; and (b) a fully reclined position, in which the backrest is disposed at a second backrest angle relative to horizontal that is shallower than the first backrest angle, the first footrest is extended in front of the seat, the seat has a second seat angle that is deeper relative to horizontal than the first seat angle, and the seat has a second longitudinal position that is rearward of the first longitudinal position. The reclining mechanism includes a footrest mechanism attached to the first footrest, the footrest mechanism configured to control relative movement of the first footrest relative to the seat. The reclining seating unit further com-

prises a linear actuator attached to the reclining mechanism and the footrest linkage and configured to move the backrest, seat and first footrest between the upright and fully reclined positions. The linear actuator includes a rail, and wherein the rail defines an angle relative to horizontal of at least 30 degrees in the upright position.

#### BRIEF DESCRIPTION OF THE FIGURES

**[0008]** FIG. 1 is a side view of a seating unit mounted on a riser arrangement according to embodiments of the invention, wherein the seating unit is shown in the upright position.

**[0009]** FIG. 2 is a side view of the seating unit of FIG. 1, shown in the fully reclined position.

**[0010]** FIG. 3 is a side view of the reclining mechanism of the seating unit of FIG. 1, shown in the upright position.

**[0011]** FIG. 4 is a side view of the reclining mechanism of the seating unit of FIG. 1, shown in the fully reclined position.

**[0012]** FIG. 5 is a top view of the seating unit of FIG. 1, shown in the upright position.

**[0013]** FIG. 6 is a top view of the reclining mechanism of the seating unit of FIG. 1, shown in the upright position.

**[0014]** FIG. 7 is a side view of a seating unit according to alternative embodiments of the invention, shown in the upright position.

**[0015]** FIG. 8 is a side view of the seating unit of FIG. 7, with the backrest in a fully reclined position, but with the footrest retracted.

**[0016]** FIG. 9 is a side view of a seating unit mounted on a riser arrangement according to additional embodiments of the invention, wherein the seating unit is shown in the upright position.

**[0017]** FIG. 10 is a side view of the seating unit of FIG. 9, shown in the fully reclined position.

**[0018]** FIG. 11 is a top view of the seating unit of FIG. 9, shown in the upright position.

**[0019]** FIG. 12 is a side view of the reclining mechanism of the seating unit of FIG. 9, shown in the upright position.

**[0020]** FIG. 13 is a side view of the reclining mechanism of the seating unit of FIG. 9, shown in the fully reclined position.

#### DETAILED DESCRIPTION

**[0021]** The present invention now is described more fully hereinafter with reference to the accompanying drawings, in which embodiments of the invention are shown. This invention may, however, be embodied in many different forms and should not be construed as limited to the embodiments set forth herein; rather, these embodiments are provided so that this disclosure will be thorough and complete, and will fully convey the scope of the invention to those skilled in the art.

**[0022]** Like numbers refer to like elements throughout. In the figures, the thickness of certain lines, layers, components, elements or features may be exaggerated for clarity. Broken lines illustrate optional features or operations unless specified otherwise.

**[0023]** The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the invention. As used herein, the singular forms “a”, “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises”

and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items. As used herein, phrases such as “between X and Y” and “between about X and Y” should be interpreted to include X and Y. As used herein, phrases such as “between about X and Y” mean “between about X and about Y.” As used herein, phrases such as “from about X to Y” mean “from about X to about Y.”

**[0024]** Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this invention belongs. It will be further understood that terms, such as those defined in commonly used dictionaries, should be interpreted as having a meaning that is consistent with their meaning in the context of the specification and relevant art and should not be interpreted in an idealized or overly formal sense unless expressly so defined herein. Well-known functions or constructions may not be described in detail for brevity and/or clarity.

**[0025]** It will be understood that when an element is referred to as being “on”, “attached” to, “connected” to, “coupled” with, “contacting”, etc., another element, it can be directly on, attached to, connected to, coupled with or contacting the other element or intervening elements may also be present. In contrast, when an element is referred to as being, for example, “directly on”, “directly attached” to, “directly connected” to, “directly coupled” with or “directly contacting” another element, there are no intervening elements present. It will also be appreciated by those of skill in the art that references to a structure or feature that is disposed “adjacent” another feature may have portions that overlap or underlie the adjacent feature.

**[0026]** The seating units illustrated and described herein comprise a plurality of pivotally interconnected links. Those skilled in this art will appreciate that the pivots between links can take a variety of configurations, such as pivot pins, rivets, bolt and nut combinations, and the like, any of which would be suitable for use with the present invention. Also, the shapes of the links may vary as desired, as may the locations of certain of the pivots. Moreover, in some instances combinations of pivot points may be replaced by equivalent structures, such as “slider-crank” configurations, like those described in B. Paul, *Kinematics and Dynamics of Planar Machinery* 4-21 (1979).

**[0027]** Referring now to the drawings, a seating unit (such as a chair) according to embodiments of the invention is illustrated in FIGS. 1-6 and designated broadly at 10. The chair 10 includes an arm frame 11 with opposed arms 12 that are connected by cross-members 14. The arm frame 11 has a cutaway area 19 that includes perpendicular front and upper edges 19a, 19b and is configured to enable the arm frame 11 to rest and be mounted on a tread 16 and riser 18 of a riser arrangement, such as may be found in a theater. The seating unit 10 also includes a seat 13 that is generally horizontally disposed above the tread 16 and includes an underlying open seat frame 22, a backrest 15 that is generally vertically and disposed substantially above a rear portion of the seat 13, and two footrests 17a, 17b, which, in the



upright position of FIG. 1, are generally vertically disposed below a front portion of the seat 13 and in front of the riser 18.

[0028] The seat 13, backrest 15 and footrests 17a, 17b are interconnected with the arm frame 11 and with each other via a pair of reclining mechanisms 30. The reclining mechanisms 30 are mirror images of each other relative to a vertical plane P (FIG. 5) that is disposed midway between the arms 12. As such, only one reclining mechanism 30 will be described herein, with the understanding that the discussion is equally applicable to the mirror image reclining mechanism 30. Also, as used herein, the terms “front,” “forward” and the like refer to a horizontal direction defined by a vector that originates in the backrest and extends parallel to the plane P toward the footrests (in other words, the forward direction is consistent with an occupant of the seating unit facing forward). The term “rear” and the like refer to the direction opposite to the forward direction. Together the front and rear directions make up the “longitudinal” dimension of the seating unit 10. The “lateral” dimension of the seating unit 10 is perpendicular to the plane P.

[0029] Further, the reclining mechanisms 30 are configured to move the seating unit 10 between an upright position, in which the backrest 15 is generally upright, the seat 13 is generally horizontal, and the footrests 17a, 17b are disposed beneath the front portion of the seat 13 (see FIGS. 1, 3, 5 and 6), and a fully reclined position, in which the backrest 15 is reclined relative to its disposition in the upright position, and the footrests 17a, 17b are disposed in front of the seat 13 (see FIGS. 2 and 4). For the sake of clarity, the reclining mechanisms 30 will first be described in the fully reclined position.

[0030] Referring now to FIGS. 2 and 4, the reclining mechanism 30 includes an angled foundation link 32 that is mounted in the inner surface of the arm 12. A spacing link 34 is fixedly mounted near the rear end of the foundation link 32. A carrier link 36 is pivotally mounted to the spacing link 34 at a pivot 37. The carrier link 36 includes a rear projection 38 that extends upwardly from the rear end of the carrier link 36 to a pivot 40 with a backpost 42 that is fixed to the backrest 15. The carrier link 36 extends forwardly, then downwardly, from the rear projection 38, to terminate in front of the riser 18.

[0031] The seat frame 22 is fixed to the carrier link 36 and extends forwardly from the rear projection 38 to terminate beneath the front portion of the seat 13. A coupling link 48 is attached at its upper end to the seat link 22 via a pin 50 that fits within an arcuate slot 22a in the seat link 22. In the fully reclined position of FIGS. 2 and 4, the pin 50 is located in the rear end of the slot 22a. The lower end of the coupling link 48 is attached to the lower end of the foundation link 32 at a pivot 52.

[0032] A connecting link 44 is attached at its rear end to the lower end of the backpost 42 at a pivot 46. The forward end of the connecting link 44 is attached to an extension 54 of a lower footrest swing link 84 (discussed further below) at a pivot 56.

[0033] The reclining mechanism 30 also includes a footrest linkage 83 with the aforementioned lower footrest swing link 84 being attached at a middle portion thereof to the seat frame 22 at a pivot 85. The lower footrest swing link 84 is also attached to the upper end of the coupling link 48 at a pivot that coincides with the pin 50. The lower footrest

swing link 84 extends forwardly from the pin 50. An upper footrest swing link 88 is also attached to the seat frame 22 at a pivot 90 and extends forwardly therefrom. An upper footrest extension link 92 is attached to the forward end of the lower footrest swing link 84 at a pivot 94 and extends upwardly and forwardly therefrom. The upper footrest extension link 92 is also attached to the upper footrest swing link 88 at a pivot 93. A lower footrest extension link 96 is attached to the forward end of the upper footrest swing link 88 at a pivot 98 and extends forwardly and upwardly therefrom. A main footrest bracket 100 is attached to the forward ends of the upper footrest extension link 92 and the lower footrest extension link 96 at, respectively, pivots 102, 104. The main footrest 17a is mounted on the main footrest bracket 100 (FIG. 2).

[0034] A lower auxiliary footrest swing link 110 is attached to the main footrest bracket 100 at the pivot 104, and an upper auxiliary footrest swing link 116 is attached to the main footrest bracket 100 at a pivot 118. An auxiliary footrest bracket 120 is attached to the front ends of the swing links 110, 116 at, respectively, pivots 122, 124. A control link 106 is attached to the forward end of the lower footrest extension link 96 at a pivot 108 and to the upper auxiliary footrest swing link 116 at a pivot 112. The auxiliary footrest 17b is mounted on the auxiliary footrest bracket 120 (FIG. 2).

[0035] Movement of the seating unit 10 between the upright position (FIGS. 1, 3, 5 and 6) and the fully reclined position (FIGS. 2 and 4) is initiated and controlled by a linear actuator 130. Referring to FIGS. 2 and 4-6, the linear actuator 130 includes a motor 132, a rail 134 that extends from the motor 132, and a carriage 136 that slides relative to the rail 134, with its position being controlled by the motor 132. The motor 132 is attached via a pivot 138 to a bracket 140 that is in turn fixed to a cross-member 142. The cross-member 142 is fixed to the lower end of the carrier link 36. The carriage 136 is fixed to an angled extension 141. The extension 141 is attached at its free end to a crank 143 at a pivot 144. The crank 143 is fixed to a bracket 146 that is fixed to the upper surface of a cross-member 148. The cross-member 148 is fixed to the underside of the lower footrest swing link 84. As such, movement of the carriage 136 of the linear actuator 130 moves the lower footrest swing link 84. In addition, a lengthy brace 152 is fixed at one end to a lower portion of the carrier link 36 above the pivot 38, and at the other end to the seat link 22 above the pivot 90.

[0036] Movement of the seating unit 10 from the upright position (FIGS. 1, 3, 5 and 6) to the fully reclined position (FIGS. 2 and 4) will now be described. Referring to FIGS. 1 and 3, in the upright position the carriage 136 of the linear actuator 130 is in its rearmost position on the rail 134. The crank 143 is positioned above the pivot 144. The lower footrest swing link 84 is oriented with the pin/pivot 50 in the forward end of the slot 22a of the seat link 22, and with the remainder of the lower footrest swing 84 extending downwardly and rearwardly therefrom. The remainder of the footrest linkage 83 of the reclining mechanism 30 is folded in a scissors-like manner under the front portion of the seat 13, with the main footrest 17a generally vertically disposed just forward of the seat 13, and the auxiliary footrest 17b generally vertically disposed rearward of the main footrest 17a. The coupling link 48 is disposed substantially parallel to the lower footrest swing link 84. The carrier link 36 is

disposed such that its forward segment **36a** is substantially vertically disposed in front of the riser **18**, and its rearward segment **36b** is substantially horizontally disposed above the tread **16**. The linear actuator **130** is oriented such that the rail **134** is tilted upwardly from the motor **132**; typically, the angle  $\alpha$  of the rail **134** relative to horizontal (e.g., relative to the tread **16**) is between about 30 and 45 degrees, and in some embodiments at least 30 degrees. As a result of the orientation of the above-described links, the seat frame **22** (and in turn the seat **13**) is substantially horizontally disposed, and the backpost **42** (and in turn the backrest **15**) is generally upright, with a slight (e.g., 10-20 degrees) rearward tilt for an occupant seated in the seating unit **10**.

**[0037]** To move the seating unit **10** from the upright position of FIGS. **1** and **3** to the fully reclined position of FIGS. **2** and **4**, the occupant of the seating unit **10** activates the linear actuator **130**. Typically the seating unit **10** will include a toggle switch, joystick, buttons, or some other actuation device that is operatively coupled to the motor **132** to induce the motor **132** to drive the carriage **136** along the rail **134**. As the carriage **136** moves forwardly and upwardly, the extension **138** causes the crank **143** to rotate (counterclockwise from the vantage point of FIG. **3**). Rotation of the crank **143** causes the bracket **146** and the cross-member **148** to rotate counterclockwise also. This rotation causes the lower footrest swing link **84** to rotate counterclockwise about the pivot **85**, which forces the pin **50** rearwardly in the slot **22a**. The rotation of the lower footrest swing link **84** drives the upper footrest extension link **92** forwardly, which in turn rotates the upper footrest swing link **88** about the pivot **90**. Rotation of the upper footrest swing link **88** drives the lower footrest extension link **96** forwardly and causes it to separate slightly from the upper footrest extension link **92**. The relative movement of the upper and lower footrest extension links **92**, **96** rotates the main ottoman bracket **100** counterclockwise to a generally horizontal position. Relative rotation of the main ottoman bracket **100** and the lower footrest extension link **96** also forces the control link **106** away from the main footrest bracket **100**, which extends the upper auxiliary footrest swing link **116** and, in turn, the auxiliary footrest bracket **120**. Extension ceases when the upper footrest swing link **88** strikes a pin **92a** on the upper footrest extension link **92**. Additional aspects of the extension of the footrests **17a**, **17b** may be discussed in U.S. Pat. No. 8,752,890 to Murphy et al., the disclosure of which is hereby incorporated herein by reference in its entirety.

**[0038]** Also, the rotation and movement of the lower footrest swing link **84** and the pin **50** attached thereto draws the coupling link **48** counterclockwise about the pivot **52** as its upper end travels with the pin **50** rearwardly in the slot **22a**. This causes the seat link **22** and seat **13** to rise slightly and take on a slightly angled pitch (the pitch angle may be between about 5 and 15 degrees). This movement draws the lower end of the segment **36a** of the carrier link **36** slightly forwardly and upwardly. Rotation of the lower footrest link **84** also draws the extension **54** forward, which movement, combined with the rising of the seat link **22**, draws the front end of the connecting link **44** forwardly and slightly upwardly. These movements pull the lower end of the backpost **42** forwardly and cause the backpost **42**, and in turn the backrest **15**, to rotate counterclockwise about the pivot **40** into a reclined disposition (typically the backrest **15** forms an angle of between about 30 and 50 degrees relative to horizontal).

**[0039]** Notably, in the upright position (see FIG. **1**) all of the links of the footrest linkage **83** and the linear actuator **130** reside below the front portion of the seat **13**. This configuration may enable the seating unit **10** to be mounted on the tread **16** and riser **18** of an existing riser arrangement without modification of the riser arrangement. In the illustrated embodiment, the footrest linkage **83** and the linear actuator **130** are able fit within a “pocket” below the seat **13** and in front of the tread **18** that is between about 10 and 16 inches in height and between about 10 and 15 inches in “depth” (measured in the longitudinal dimension). This capability is facilitated by the upward angle at which the linear actuator **130** is mounted in the upright position. This capability may also be facilitated by the movement of the entire linear actuator **130** (including the motor **132**) with the carrier link **36** when moving from the upright to the fully reclined position.

**[0040]** Also, the seating unit **10** has the capability of reaching the fully reclined position (in which the backrest **15** is reclined to between about 30 and 50 degrees) while still providing a reasonable seating height (e.g., between about 17 and 23 inches above the floor surface below the front portion of the seat **13**). This is facilitated by the carrier link **36**, which both underlies the rear portion of the seat **13** and provides a low mounting point for the linear actuator **130**. This capability is also facilitated by the coupling of the coupling link **48** with the lower footrest swing link **84** via the pin **50** and the rearward travel of this pivot point within the slot **22b** on the seat frame **22**.

**[0041]** Those of skill in this art will appreciate that the seating unit **10** may take other forms. For example, the seating unit **10** may include more than one linear actuator, wherein one actuator controls the reclining of the backrest and the other controls the extension of the footrest(s). One example of such a design is shown in FIGS. **7** and **8** and designated broadly at **210**. The seating unit **210** includes a footrest linkage **283** that is essentially the same as the footrest linkage **83** discussed above. The seating unit **210** also includes a reclining mechanism **230** that is largely similar to the reclining mechanism **30** discussed above. However, the seating unit **210** includes a second linear actuator **400** that allows the backrest **215** to recline independently of the extension/retraction of the footrests **217a**, **217b**. The differences in the reclining mechanisms **30**, **230** are discussed below.

**[0042]** Referring first to FIG. **7**, the reclining mechanism **230** includes a carrier link **236** that with a projection **238** that is smaller than the projection **38**. A two-piece backpost **242** is pivotally attached to the projection **238** at a pivot **240** and also engages the projection **238** via a pin **240a** that is received in a slot **238a** in the projection **238**. Another projection **244** is fixed to and extends rearwardly from the carrier link **236** and the projection **238**. A cross-member **246** extends between the projections **244** on either side of the seating unit **210**. A bracket **248** is fixedly mounted on the cross-member **246**. The linear actuator **400** is mounted at a pivot **250** to the bracket **248** and extends upwardly therefrom to a pivot **254** with a bracket **252**. The bracket **252** is mounted to a cross-member **256** that extends between and is fixed to the upper ends of the backposts **242** on either side of the seating unit **210**. The backrest **215** is fixed to the backpost **242**.

**[0043]** In the upright position (FIG. **7**), the linear actuator **400** is fully extended, and the pin **242a** is positioned in the

rear end of the slot 238a. In this position the backrest 215 is relatively upright. When an occupant of the seating unit 210 wishes to recline the backrest 215, the occupant activates the linear actuator 400 to retract. This movement draws the upper end of the backpost 242 downwardly, which in turn causes the backpost 242 to rotate counterclockwise about the pivot 250. This action of the backpost 242 draws the backrest 215 into a more reclined position (see FIG. 8). The backrest 215 is fully reclined when the pin 242a reaches the forward end of the slot 238a. Thus, it can be seen that the backrest 215 of the seating unit 210 can be reclined without extending the footrests 217a, 217b. (Conversely, the footrests 217a, 217b can be extended in front of the seat 214 in the manner described above using the linear actuator 300 without reclining the backrest 215).

[0044] Referring now to FIGS. 9-13, another seating unit, designated broadly at 500, is shown therein. The seating unit 500 includes reclining mechanisms 530 that are similar, but not identical, to the reclining mechanisms 30 discussed above. Differences in the reclining mechanisms 530 are discussed below.

[0045] Referring now to FIGS. 10 and 13 (which show the reclining mechanism 530 in the fully reclined position), the reclining mechanism 530 includes an angled foundation link 532 that is mounted in the inner surface of the arm 512. An angled link 533 is attached at its forward end to the foundation link 532 at a pivot 535. A control link 537 is attached to the rear end of the angled link 533 at a pivot 539. A carrier link 536 is pivotally mounted to the vertex of the angled link 533 at a pivot 541. The carrier link 536 includes a rear projection 538 that extends upwardly from the rear end of the carrier link 536 to a pivot 540 with a backpost 542 that is fixed to the backrest 515. The carrier link 536 extends forwardly, then downwardly, from the rear projection 538, to terminate in front of the riser 518. The lower end of the backpost 542 is pivotally attached to the upper end of the control link 537 at a pivot 546.

[0046] The seat link 522 is fixed to the carrier link 536 and extends forwardly from the rear projection 538 to terminate beneath the front portion of the seat 513. A coupling link 548 is attached at its upper end to the seat link 522 via a pivot 550. The lower end of the coupling link 548 is attached to the lower end of the foundation link 532 at a pivot 552. A tripartite link 560 is attached at its lower end to the carrier link 536 at a pivot 562. At its upper end, the tripartite link 560 is attached to the forward end of the seat link 522 at a pivot 564. An extension panel 568 extends downwardly from the front end of the seat link 522.

[0047] A connecting link 544 is attached at its rear end to a forward portion of the foundation link 532 at a pivot 551. The forward end of the connecting link 544 is attached to an upper footrest swing link 588 (discussed further below) at a pivot 556.

[0048] The reclining mechanism 530 also includes a footrest linkage 583. A lower footrest swing link 584 is attached at its rear end to the carrier link 536 at a pivot 632 and extends forwardly therefrom. The aforementioned upper footrest swing link 588 is attached to the seat link 522 at a pivot 590 and extends forwardly therefrom. An upper footrest extension link 592 is attached to the forward end of the lower footrest swing link 584 at a pivot 594 and extends upwardly and forwardly therefrom. The upper footrest extension link 592 is also attached to the upper footrest swing link 588 at a pivot 593. A lower footrest extension link

596 is attached to the forward end of the upper footrest swing link 588 at a pivot 598 and extends forwardly and upwardly therefrom. A main footrest bracket 600 is attached to the forward ends of the upper footrest extension link 592 and the lower footrest extension link 596 at, respectively, pivots 602, 604. The main footrest 517a is mounted on the main footrest bracket 600 (FIG. 10).

[0049] A lower auxiliary footrest swing link 610 is attached to the main footrest bracket 600 at the pivot 604, and an upper auxiliary footrest swing link 616 is attached to the main footrest bracket 600 at a pivot 618. An auxiliary footrest bracket 620 is attached to the front ends of the swing links 610, 616 at, respectively, pivots 622, 624. A control link 606 is attached to the forward end of the lower footrest extension link 596 at a pivot 608 and to the upper auxiliary footrest swing link 616 at a pivot 612. The auxiliary footrest 517b is mounted on the auxiliary footrest bracket 620 (FIG. 10).

[0050] Movement of the seating unit 510 between the upright position (FIGS. 9, 11 and 12) and the fully reclined position (FIGS. 10 and 13) is initiated and controlled by a linear actuator 630, which is in turn attached to an actuating linkage 631. Referring to FIGS. 10, 11 and 13, the linear actuator 630 includes a motor 632, a rail 634 that extends from the motor 632, and a carriage 636 that slides relative to the rail 634, with its position being controlled by the motor 632. The motor 632 is attached via a pivot 638 to a bracket 640 that is in turn fixed to a cross-member 642. The cross-member 642 is fixed to the lower end of the carrier link 536. The carriage 636 is fixed to an angled extension 641. The extension 641 is attached at its free end to a crank 643 at a pivot 644. The crank 643 is fixed to a bracket 646 that is fixed to the upper surface of a cross-member 648. The cross-member 648 is fixed to the underside of the lower footrest swing link 584. As such, movement of the carriage 636 of the linear actuator 630 moves the lower footrest swing link 584.

[0051] Movement of the seating unit 10 from the upright position (FIGS. 9, 11 and 12) to the fully reclined position (FIGS. 10 and 13) will now be described. Referring to FIGS. 9 and 12, in the upright position the carriage 636 of the linear actuator 630 is in its lowest position on the rail 634. The crank 643 is positioned above and rearward of the pivot 644. The tripartite link 560 extends upwardly and forwardly from the pivot 562. The lower footrest swing link 584 extends downwardly and rearwardly from the pivot 632 with the carrier link 536. The remainder of the footrest linkage 583 of the reclining mechanism 520 is folded in a scissors-like manner under the front portion of the seat 513, with the main footrest 517a generally vertically disposed just forward of the seat 513, and the auxiliary footrest 517b generally vertically disposed rearward of the main footrest 517a. The coupling link 548 extends downwardly and rearwardly from the seat link 522. The carrier link 536 is disposed such that its forward segment 536a is substantially vertically disposed in front of the riser 518, and its rearward segment 536b is substantially horizontally disposed above the tread 16. The linear actuator 630 is oriented such that the rail 634 is tilted upwardly from the motor 632; typically, the angle  $\alpha$  of the rail 634 relative to horizontal (e.g., relative to the tread 16) is between about 45 and 80 degrees, and in some embodiments at least 50 degrees. As a result of the orientation of the above-described links, the seat link 522 (and in turn the seat 513) is substantially horizontally disposed, and the backpost

**542** (and in turn the backrest **515**) is generally upright, with a slight (e.g., 10-20 degrees) rearward tilt for an occupant seated in the seating unit **510**.

**[0052]** To move the seating unit **10** from the upright position of FIGS. **9** and **11** and **12** to the fully reclined position of FIGS. **10** and **13**, the occupant of the seating unit **510** activates the linear actuator **630**. Typically the seating unit **510** will include a toggle switch, joystick, buttons, or some other actuation device that is operatively coupled to the motor **632** to induce the motor **632** to drive the carriage **636** along the rail **634**. As the carriage **636** moves forwardly and upwardly, the extension **641** causes the crank **643** to rotate (counterclockwise from the vantage point of FIG. **12**). Rotation of the crank **643** causes the bracket **646** and the cross-member **648** to rotate counterclockwise also. This rotation causes the lower footrest swing link **584** to rotate counterclockwise about the pivot **632**. The rotation of the lower footrest swing link **584** drives the upper footrest extension link **592** forwardly, which in turn rotates the upper footrest swing link **588** about the pivot **590**. Rotation of the upper footrest swing link **588** drives the lower footrest extension link **596** forwardly and causes it to separate slightly from the upper footrest extension link **592**. The relative movement of the upper and lower footrest extension links **592**, **596** rotates the main ottoman bracket **600** counterclockwise to a generally horizontal position. Relative rotation of the main ottoman bracket **600** and the lower footrest extension link **596** also forces the control link **606** away from the main footrest bracket **600**, which extends the upper auxiliary footrest swing link **616** and, in turn, the auxiliary footrest bracket **620**. Extension ceases when the upper footrest swing link **588** strikes a pin on the upper footrest extension link **592**. Additional aspects of the extension of the footrests **517a**, **517b** may be discussed in U.S. Pat. No. 8,752,890 to Murphy et al., the disclosure of which is hereby incorporated herein by reference in its entirety.

**[0053]** Also, the rotation and movement of the lower footrest swing link **584** draws the connecting link **544** counterclockwise about the pivot **551**. This causes the seat link **522** and seat **513** to rise slightly and take on a slightly angled pitch (the pitch angle may be between about 5 and 15 degrees). The movement also causes the seat link **522** and seat **513** to move rearwardly (in some embodiments the rearward movement is between about 2 and 4.5 inches). Movement of the seat link **522** and seat **513** is also controlled by the tripartite link **560** and the coupling link **548**, which rotate counterclockwise about the pivots **562**, **552**, respectively. The movement of the seat link **522** rotates the carrier link **536** counterclockwise and moves it rearwardly. The motion of the carrier link **536** drives the backpost **542** (and in turn the backrest **515**) to rotate counterclockwise, with its movements controlled by the angled link **533** and the control link **537**. Typically the backrest **515** forms an angle of between about 30 and 50 degrees relative to horizontal).

**[0054]** Notably, the rearward movement of the seat **522** results in the footrests **517a**, **517b** being disposed a shorter distance in front of the riser **518** (e.g., 2 to 4.5 inches) while still being disposed the same distance in front of the seat **522** as with the reclining mechanism **30**. As a result, more room is available in front of the footrest **517b** when the seating unit **510** is in the fully reclined position, which may provide additional room for someone to pass in front of the extended footrest **517b**.

**[0055]** As another exemplary variation, the seating unit **10** may also be constructed such that, when reclined, the angle between the seat and backrest does not change. Further, the type of linear actuator employed may vary (e.g., a telescoping design, in which a rod retracts and extends, may also be used). Other variations may also be apparent to those of skill in this art.

**[0056]** It should also be noted that, although the seating unit is illustrated for use with a riser arrangement, such as that employed in a theater, the seating units **10**, **310**, **510** may be used in other settings. For example, in some instances the arms may lack a cutaway area, and the reclining mechanism may be used with a conventional chair. Such a chair may be particularly suitable for use in a health care environment.

**[0057]** Further, it should be apparent that the occupant may choose to cease movement of the seating unit at any point between the upright and fully reclined positions by simply halting the movement of the carriage of the linear actuator on the rail.

**[0058]** The foregoing is illustrative of the present invention and is not to be construed as limiting thereof. Although exemplary embodiments of this invention have been described, those skilled in the art will readily appreciate that many modifications are possible in the exemplary embodiments without materially departing from the novel teachings and advantages of this invention. Accordingly, all such modifications are intended to be included within the scope of this invention as defined in the claims. The invention is defined by the following claims, with equivalents of the claims to be included therein.

What is claimed is:

1. A reclining seating unit, comprising:

a frame having a pair of arms, wherein each of the arms includes a cutaway portion in a lower rear area thereof, the cutaway portion configured to receive a tread and a riser of a riser arrangement;

a backrest;

a seat;

a first footrest;

a reclining mechanism connected between the backrest, seat, and first footrest, the reclining mechanism comprising a series of pivotally interconnected links and configured to move the seating unit between: (a) an upright position, in which the backrest is disposed at a first generally upright backrest angle relative to the seat, the seat is disposed at a first generally horizontal seat angle and at a first longitudinal position, and the first footrest is retracted below a forward portion of the seat; and (b) a fully reclined position, in which the backrest is disposed at a second backrest angle relative to horizontal that is shallower than the first backrest angle, the first footrest is extended in front of the seat, the seat has a second seat angle that is deeper relative to horizontal than the first seat angle, and the seat has a second longitudinal position that is rearward of the first longitudinal position;

wherein the reclining mechanism includes a footrest mechanism attached to the first footrest, the footrest mechanism configured to control relative movement of the first footrest relative to the seat;

- a linear actuator attached to the reclining mechanism and the footrest linkage and configured to move the backrest, seat and first footrest between the upright and fully reclined positions;
- wherein the linear actuator and the footrest linkage reside entirely below a front portion of the seat and in front of the cutaway portion of the arms.
2. The seating unit defined in claim 1, wherein the cutaway portion is defined by a front edge and an upper edge, and wherein the front edge is substantially perpendicular to the upper edge.
3. The seating unit defined in claim 1, wherein the linear actuator includes a rail, and wherein the rail is disposed at an angle of at least 30 degrees in the upright position.
4. The seating unit defined in claim 1, further comprising a second footrest, wherein in the upright position, the second footrest is generally vertically disposed and rearward of the first footrest, and in the fully reclined position the second footrest is generally horizontally disposed and forward of the first footrest.
5. The seating unit defined in claim 1, wherein the reclining mechanism further includes a carrier link that is fixed relative to the seat, and wherein the linear actuator is pivotally attached to a lower forward end portion of the carrier link.
6. The seating unit defined in claim 5, wherein the backrest is fixed to a backpost, and wherein the carrier link is pivotally connected to the backpost.
7. A reclining seating unit, comprising:
- a frame having a pair of arms;
  - a backrest;
  - a seat;
  - a first footrest;
  - a reclining mechanism connected between the backrest, seat, and first footrest, the reclining mechanism comprising a series of pivotally interconnected links and configured to move the seating unit between: (a) an upright position, in which the backrest is disposed at a

- first generally upright backrest angle relative to the seat, the seat is disposed at a first generally horizontal seat angle and at a first longitudinal position, and the first footrest is retracted below a forward portion of the seat; and (b) a fully reclined position, in which the backrest is disposed at a second backrest angle relative to horizontal that is shallower than the first backrest angle, the first footrest is extended in front of the seat, the seat has a second seat angle that is deeper relative to horizontal than the first seat angle, and the seat has a second longitudinal position that is rearward of the first longitudinal position;
- wherein the reclining mechanism includes a footrest mechanism attached to the first footrest, the footrest mechanism configured to control relative movement of the first footrest relative to the seat;
- a linear actuator attached to the reclining mechanism and the footrest linkage and configured to move the backrest, seat and first footrest between the upright and fully reclined positions;
- wherein the linear actuator includes a rail, and wherein the rail defines an angle relative to horizontal of at least 30 degrees in the upright position.
8. The seating unit defined in claim 7, further comprising a second footrest, wherein in the upright position, the second footrest is generally vertically disposed and rearward of the first footrest, and in the fully reclined position the second footrest is generally horizontally disposed and forward of the first footrest.
9. The seating unit defined in claim 7, wherein the reclining mechanism further includes a carrier link that is fixed relative to the seat, and wherein the linear actuator is pivotally attached to a lower forward end portion of the carrier link.
10. The seating unit defined in claim 7, wherein the backrest is fixed to a backpost, and wherein the carrier link is pivotally connected to the backpost.

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